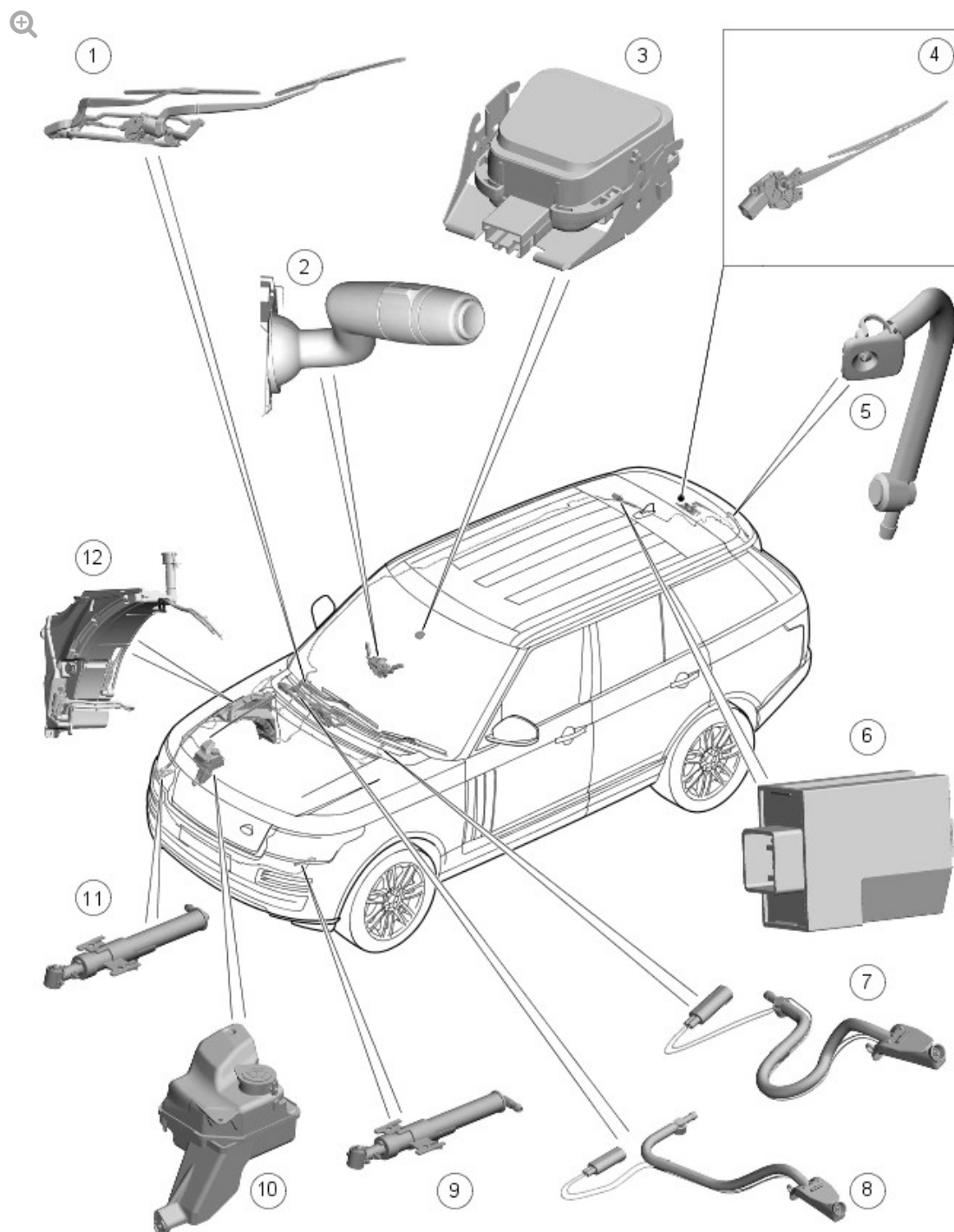


2016.0 RANGE ROVER (LG), 501-16

WIPERS AND WASHERS

DESCRIPTION AND OPERATION

COMPONENT LOCATION



E149109

ITEM	DESCRIPTION
1	Windshield wiper assembly
2	Wiper control switch
3	Rain/Light sensor
4	Rear window wiper
5	Rear window washer jet

6	RF filter
7	Left heated washer jet
8	Right heated washer jet
9	Left heated washer jet
10	Windshield washer reservoir (cold climate markets only)
11	Right heated washer jet
12	Windshield washer reservoir

OVERVIEW

The wipers and washers system functions on receipt of requests made by the driver or the rain/light sensor. All wiper functions for the windshield and rear window wipers are controlled from a wiper control switch located on the right hand side of the steering column.

The wiper and washer system comprises:

- Windshield and rear window wiper motors
- A windshield wiper linkage
- Two windshield and one rear window wiper arms and blades
- Two windshield heated washer jets and one rear window washer jet
- Two headlamp washer jets
- A windshield washer reservoir, a windshield/rear window washer pump and a headlamp washer pump
- A wiper control switch
- A rain/light sensor.

The wiper system is equipped with a rain/light sensor, which controls the automatic wiper functions and also the automatic lighting functions. The sensor, located below the interior rear view mirror, detects rain drops on the windshield and automatically operates the wipers in the intermittent mode.

The windshield wiper control switch must be in the intermittent position for rain sensor controlled wiper operation.

The windshield wiper system has 4 wiper stages of operation and 6 intermittent wiper delay periods. The 4 wiper stages are as follows:

- Flick wipe
- Intermittent
- Normal (slow) speed continuous
- Fast speed continuous.

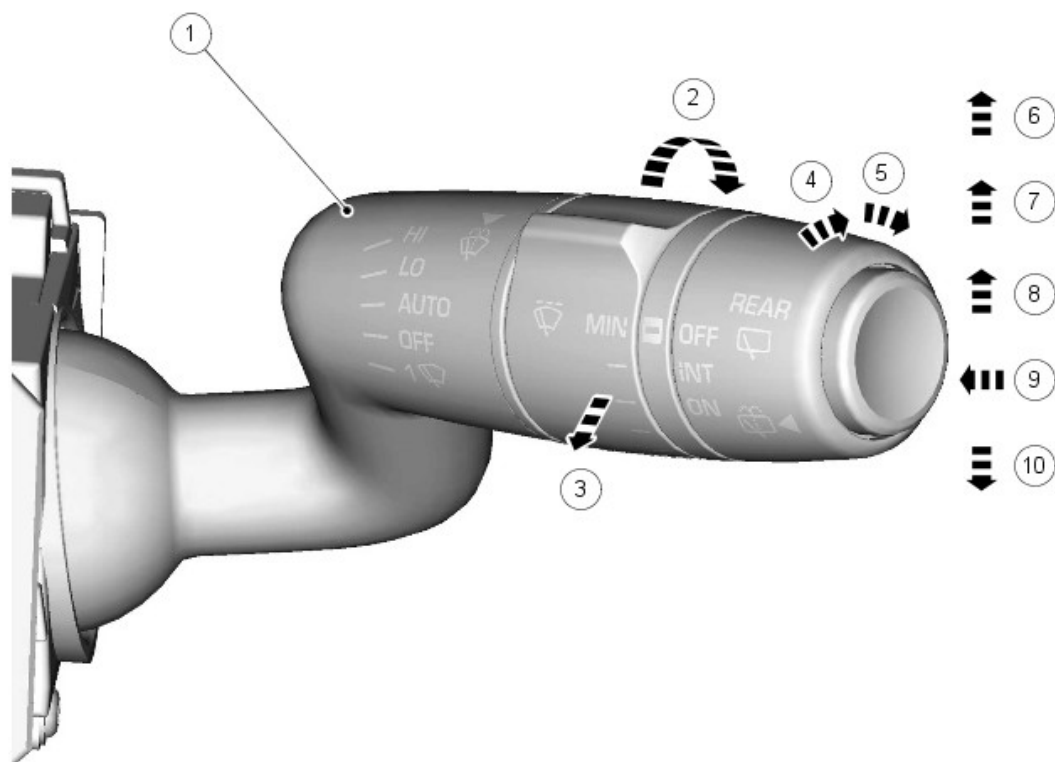
The intermittent, normal and fast speeds are affected by road speed. The intermittent wiper delay periods change with the road speed, with the delay decreasing as the road speed increases. The normal continuous operation changes to intermittent operation when the vehicle is stationary.

For the wipers and washers to operate the ignition must be on. Wiper functions are suspended during engine cranking to reduce battery power consumption under high load conditions.

Diagnostic information for the wiper system is available and can be retrieved using the Land Rover approved diagnostic equipment.

DESCRIPTION

WINDSHIELD WIPER CONTROL SWITCH



E149110

ITEM	DESCRIPTION
1	Windshield wiper control switch
2	Intermittent delay control or rain/light sensor sensitivity control
3	Programmed wash/wipe position
4	Rear window wiper intermittent delay position
5	Rear window wiper normal wipe position
6	Fast wipe position
7	Normal wipe position
8	Intermittent wipe position or rain/light sensor position
9	Rear window wash/wipe position
10	Flick wipe position

The windshield wiper control switch is located on the right hand side of the steering column and controls all windshield and rear window wiper functions.

Moving the switch up 1 position selects intermittent windshield wiper operation. The rotary potentiometer on the stalk selects 1 of 6 delay periods. When a rain/light sensor is incorporated into the system, the intermittent position also initiates wiper operation controlled by the rain/light sensor. The sensitivity of the rain/light sensor can be adjusted by rotating the rotary potentiometer to 1 of the 6 positions. Moving the switch up 1 further position selects normal (slow) continuous wiper operation. Moving the switch up to the last position selects fast continuous wiper operation.

Moving the switch down selects the flick wipe function. The windshield wipers will operate at normal speed for as long as the flick wipe switch position is operated and will stop at the park position when the switch is released.

Moving the switch rearwards, towards the driver, selects the windshield programmed wash/wipe function. The washer pump will operate for as long as the switch is held. A short operation operates the pump momentarily and the wipers complete 3 full cycles before stopping. The headlamp washers will operate if programmed wash/wipe is selected and the headlamps are on.

Rotating the outer ring of the wiper control switch to the first position selects the rear window wiper intermittent mode on. Rotating the outer ring of the wiper control switch further on to the second position selects the rear window wiper normal mode on. In intermittent mode, the rear window wiper will complete 1 full cycle and will then operate on an intermittent function until selected off. The delay between wipes decreases as vehicle speed increases. When the rear window wash/wipe switch is pressed momentarily, the rear window wiper completes 5 full cycles with 3

automatic, short operations of the rear window washer pump.

The rear window wiper also operates continuously if reverse gear is selected and the windshield wipers are on.

The windshield wiper control switch comprises 8 switch positions and two rotary controls. The inner rotary control selects the intermittent delay between wipes. The outer rotary control selects the rear window wiper positions.

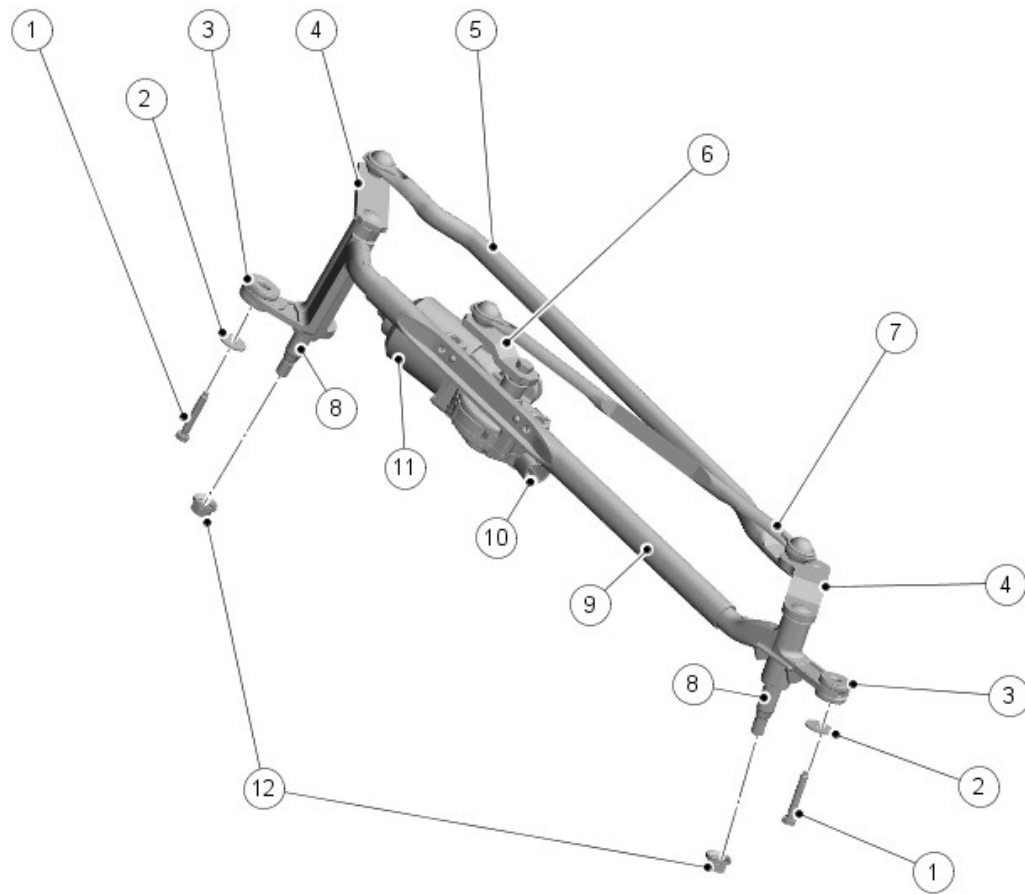
The switch positions each complete a combination of earth paths in circuits connected to the control module integrated into the clockspring. The control module interprets the switch positions and transmits the information on the LIN (local interconnect network) bus connected to the CJB (central junction box), which then operates the selected function accordingly.

WINDSHIELD WIPERS

The windshield wiper system comprises:

- Wiper motor and linkage assembly
- Washer reservoir and pumps
- Wiper arms and blades.

WIPER LINKAGE



E149111

ITEM	DESCRIPTION
1	Bolt (2 off)
2	Washer (2 off)
3	Bush (2 off)
4	Crank (2 off)
5	Tube
6	Motor crank
7	Link rod
8	Pivot housing assembly
9	Tube
10	Electrical connector

11	Windshield wiper motor
12	Self-locking nut (2 off)

The wiper linkage and motor assembly are available as separate components. The wiper linkage differs between LHD (left-hand drive) and RHD (right-hand drive) models. The wiper motor is common to LHD and RHD drive models.

The assembly is located below the plenum grill in the engine compartment and is secured with bushes, sleeves and bolts. The rubber bushes isolate the assembly from the body mountings.

The linkage assembly comprises a main tube, with a pivot housing at each end. A bracket is attached, offset along the tube length, which locates the motor assembly. A motor crank is positively attached to the motor output shaft. A link rod is connected to the motor crank and is connected at the opposite end to the right pivot housing, via a crank. The right link rod has a pivot attachment for the left link rod, which is connected at its opposite end to the right pivot housing, via a second crank.

The motor crank converts rotary motion from the motor output shaft into linear movement of the link rods. The cranks, connected between the each link rod and pivot housing, convert the linear motion back to rotary motion of the pivot housing. This rotary motion is passed to the wiper arms and blades causing the blades to wipe an arc across the windshield.

WIPER MOTOR

The motor assembly comprises a DC (direct current) motor, which drives a gear wheel via a worm drive attached to the motor spindle. The gear wheel has a central spigot, which provides the attachment point for the motor crank.

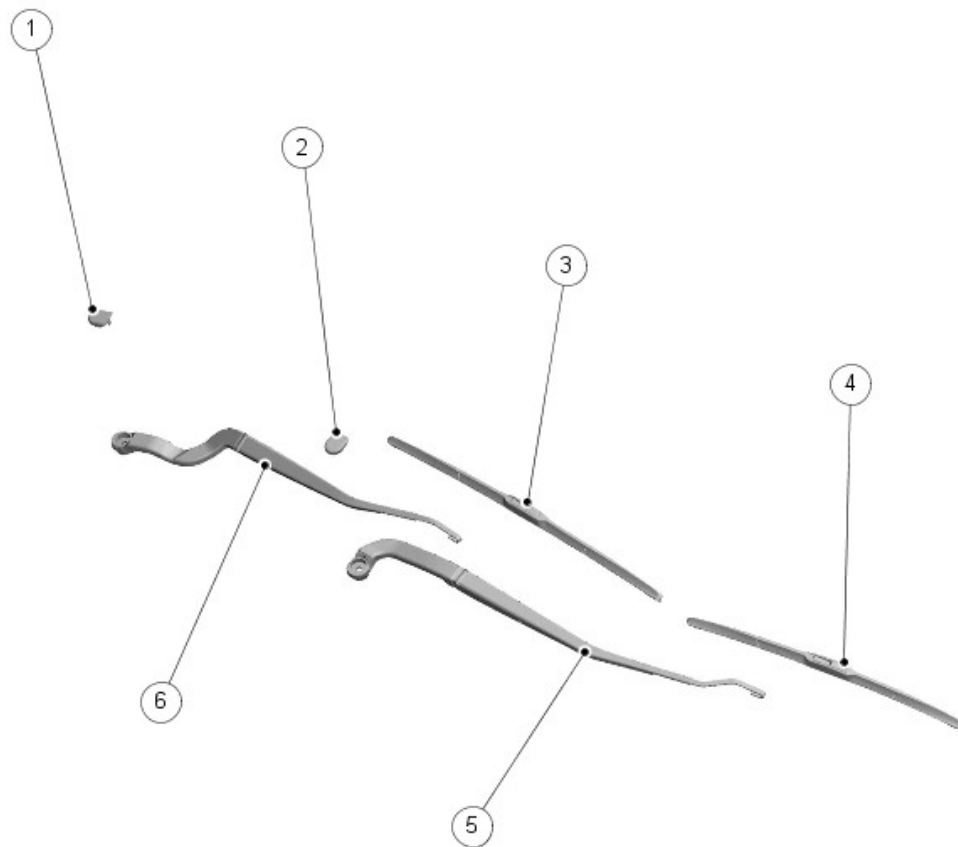
The motor is connected electrically by a four-pin connector. The connector supplies 2 battery voltage feeds to the motor. The motor has 3 sets of brushes with 1 brush connected to ground. One feed is direct to the motor

brush opposite the ground brush and operates the motor at normal (slow) speed. The second feed is connected to a motor brush, which is offset from the ground brush and operates the motor at fast speed. With the power supplied through this brush, the current flows through fewer coil windings. This results in a lower resistance to the current flow to the ground brush and gives a higher motor rotational speed.

Output control of the wiper motor is through a double contact relay. The relay is located in the EJB (engine junction box).

The motor has an internal track, which signals the CJB when the wipers have reached the park position. The park signal is an open circuit when the wipers are in the park position. When the wipers are switched off and the CJB receives the park position signal from the motor, the CJB shorts the motor via a relay bridge circuit. This short circuit has the effect of applying a brake to the motor, giving precise positioning of the wiper blades in the park position.

WIPER ARMS



E149112

ITEM	DESCRIPTION
1	Self-locking nut cover
2	Self-locking nut cover
3	Wiper blade
4	Wiper blade
5	Wiper arm
6	Wiper arm

The wiper arms are positively located on tapered splines on the wiper linkage spindles. The arm is located on a curved wiper angle adjuster, which mates with the curved underside of the arm attachment. A curved slot in the

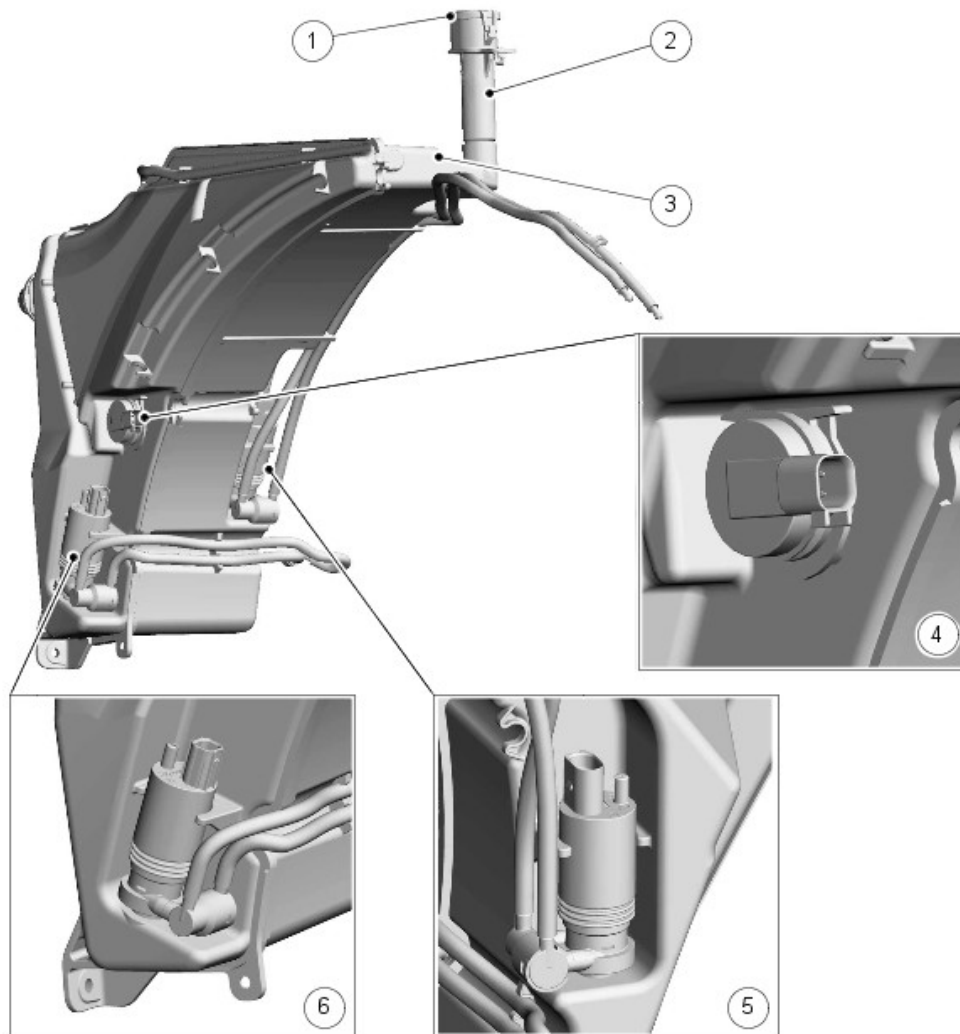
arm attachment locates an adjustment shim and the 3 components are retained under compression on the spindle spline by a self-locking nut.

The wiper angle adjuster and the angle adjust shim allow the approach angle of the wiper blade to be finely adjusted to its optimum wiping angle. The adjustment allows the approach angle of the blade to be adjusted to 2° on each side of the spindle axis. This adjustment provides precise adjustment of the blade approach angle for maximum wiper performance. The wiper angle adjuster is pressed onto the spindle and requires a tool for removal. Once removed, the wiper angle adjuster must be discarded and new one fitted on re-assembly.

The wiper arm has a pivot point, midway between the spindle attachment and the blade. Two tension springs are connected to the wiper arm on each side of the pivot point and apply pressure to maintain the wiper blade in contact with the windshield.

The wiper blades are attached to the wiper arms with clips that allow the blade to pivot. Each blade comprises a number of levers and yokes to which the rubber wiper blade is attached. The levers and yokes ensure that the pressure applied by the arm tension spring is distributed evenly along the full length of the blade and also allow the blade to adjust to the curvature and contour of the windshield. The driver's side blade is fitted with an aerofoil, which presses the blade onto the windshield at high speed, improving the wiper performance.

WASHER RESERVOIR AND PUMPS



E149113

ITEM	DESCRIPTION
1	Windshield washer reservoir filler cap
2	Windshield washer reservoir filler neck
3	Windshield washer reservoir
4	Washer fluid level sensor
5	Headlamp washer pump
6	Windshield/Rear window washer pump

The windshield washer system comprises a washer fluid reservoir, a washer pump, 2 washer jets and hoses.

The plastic, molded reservoir is located in the right wheel arch, behind the

liner and has a capacity of 12.3 pints (7 litres). It is secured with bolts and washers.

The reservoir has 2 recessed holes on its rear face, which provide location for the windshield/rear window and headlamp washer pumps. The pumps are push fitted into grommets, which seal the pumps in their locations. A hole in the top of the reservoir allows for the fitment of a flexible filler tube.

A hole in the base of the reservoir provides the location for the fluid level sensor. The hole provides access to an internal tube inside the reservoir, into which the sensor is fitted. The sensor has 2 pegs, which positively locate the sensor in the reservoir when turned through 90°.

The sensor reacts to the influence of a magnetic field. A float, with integral magnet, is located around the tube, inside the reservoir. The sensor has 2 contacts inside a glass tube, which are normally open. When the fluid level reduces, the magnetic float moves down the tube. When the magnet reaches the sensor contacts, the magnetic field closes the contacts. This completed circuit is sensed by the CJB, which displays the low fluid level message in the IC.

HEATED WASHER JETS

Two windshield washer jets are located in the finisher at the rear of the hood. The washer fluid feed hose from the windshield pump is connected to a 'Y' piece connector located between the 2 jets. Two short lengths of hose connect the jets to the 'Y' piece. Each jet contains a Non-Return Valve (NRV) to prevent washer fluid draining back to the reservoir and also to limit the amount of washer fluid, which can be forced by gravity from the jet during cornering. Each washer jet has a ball nozzle that can rotate in the housing to obtain the optimum fluid application onto the windshield.

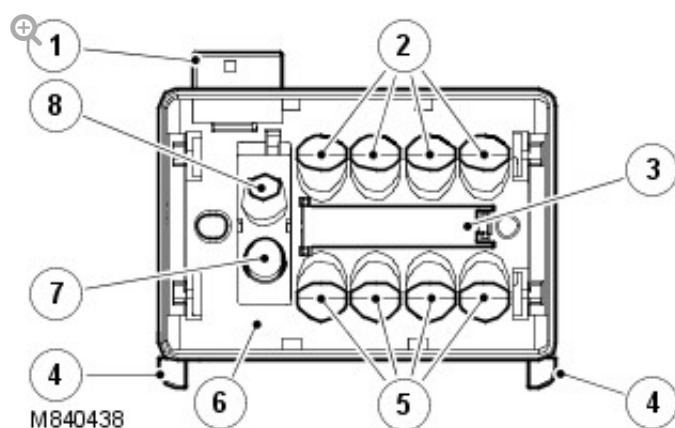
Each washer jet contains a heater element, which prevents the fluid freezing in the nozzles in very cold conditions. The jet heater elements are controlled by the ATCM (automatic temperature control module), which operates a heated washer jets relay in the EJB to energize the heater elements when the ambient air temperature falls below a predetermined temperature.

WIPER BLADE HEATING

The obscuration band along the lower edge of the windshield contains 6 heating elements that run from one side of the windshield to the other. These elements are designed to heat the area of the windshield where the wiper blades park, to prevent freezing in cold conditions.

The blade heating elements are automatically controlled by the ATCM and are activated simultaneously with the heated washer jets.

RAIN/LIGHT SENSOR



ITEM	DESCRIPTION
1	Connector
2	Receiver diodes
3	Heater element
4	Retaining clip
5	Transmitter diodes
6	Rain/light sensor body
7	Ambient light sensor
8	Forward light sensor

The rain/light sensor is located at the upper edge of the windshield, behind the interior rear view mirror. The sensor is mounted on an optical unit which is heat bonded to the inner surface of the windshield during manufacture. If

damage occurs to the optical unit or the windshield, then a new windshield will be required and fitment can only be performed by an authorized Land Rover dealer.

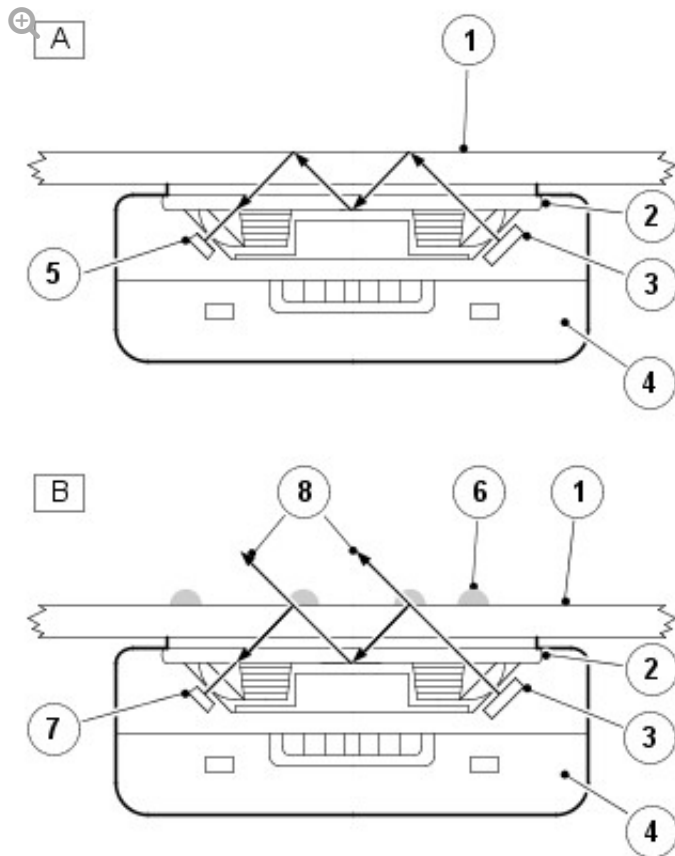
The rain/light sensor unit attaches to the optical unit via 4 clips which latch onto formed tags on the optical unit. Positive retention is achieved by 2 retaining clips which force the clips onto the tags. The retaining clips must be withdrawn to facilitate sensor removal.

The sensor provides information to the CJB, via the LIN bus, for the optimum wiper operation for the prevailing conditions to maintain the screen in a clear condition at all times. The rain/light sensor is an optical unit, which operates on an infrared waveband. The sensor uses the principle of the laws of reflection on interfacing surfaces between materials with differing refraction indices.

The sensor contains 4 transmitter and receiver diodes for increased sensitivity and also contains a light sensor for operation of the automatic headlamp function.

For additional information, refer to: [Exterior Lighting](#) (417-01 Exterior Lighting, Description and Operation).

RAIN/LIGHT SENSOR FUNCTIONALITY



M840421

ITEM	DESCRIPTION
1	Windshield - outside surface
2	Optical unit
3	Transmitter diodes (100% light transmitted)
4	Rain/light sensor unit
5	Receiver diodes (100% received)
6	Water droplets/film
7	Receiver diodes (less than 100% light received)
8	Lost light

The rain/light sensor contains transmitter and receiver diodes, which transmit and receive infrared light, which is directed onto the windshield via an optical unit. The light is directed at an angle so that the light is reflected 100% on the outside surface of the screen and is transmitted back into the

optical unit. To receive a 100% reflection, the outer screen surface must be clean and dry.

The light is reflected 4 times from when it leaves the transmitter diodes to when it is picked up by the receiver diodes. If the windshield is wet or dirty in the area of the optical unit, the clean conditions for 100% reflection means that some of the light reflected is lost. As the screen becomes dirtier or wetter, the received light is evaluated by the rain/light sensor and translated into a signal value. A micro-controller within the sensor monitors the change in signal and initiates the appropriate wipe cycle via LIN bus signals to the CJB.

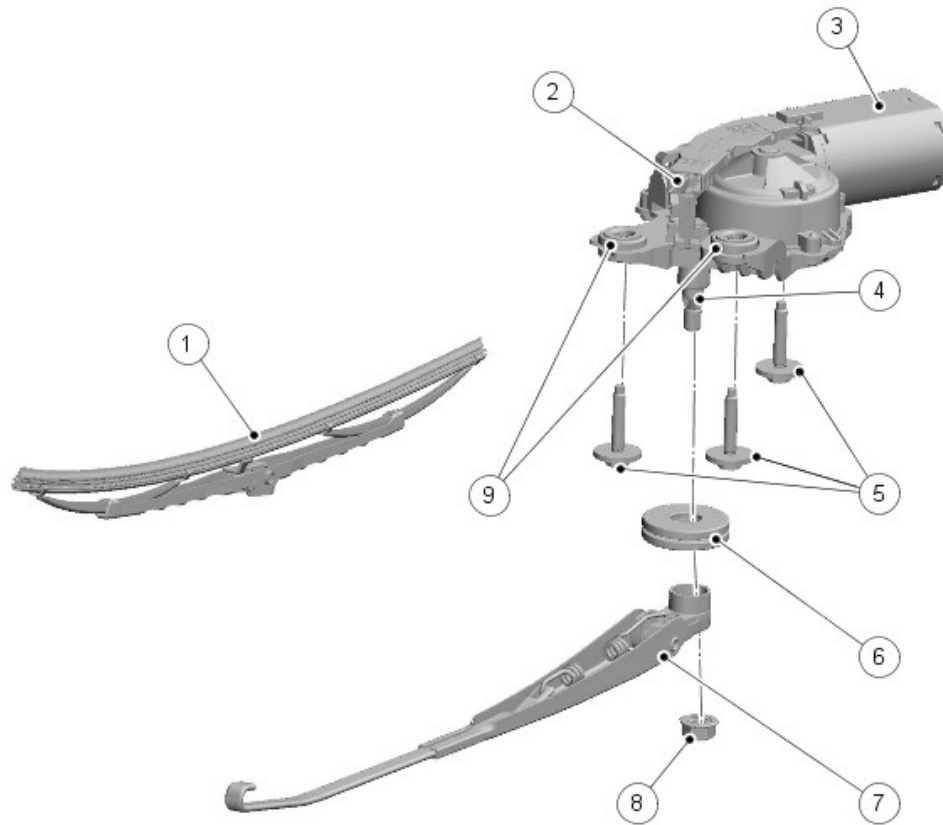
The software can compensate for the long-term effects of scratches and stone chips in the area of the optical unit and the short term effects of dirt or smears caused by worn wiper blades. A heater element is also contained within the rain/light sensor and uses ambient air temperature LIN bus signals from the CJB to keep the optical unit clear of frost or condensation.

The rain/light sensor incorporates a light guide. The light guide directs the ambient light and a proportion of the forward light (from the driving direction) to a light sensitive diode. This is used for the rain sensor sensitivity in low ambient light levels and is also used to control the automatic headlamp function.

For additional information, refer to: [Exterior Lighting](#) (417-01 Exterior Lighting, Description and Operation).

Intermittent (single wipe), continuous slow, and continuous fast windshield wiper operations are controlled by the rain/light sensor when enabled.

REAR WINDOW WIPER



E149114

ITEM	DESCRIPTION
1	Rear window wiper blade
2	Electrical connector
3	Rear window wiper motor
4	Pivot housing assembly
5	Bolt (3 off)
6	Bush
7	Rear window wiper arm
8	Self-locking nut
9	Bush (3 off)

The rear window wiper system comprises:

- Wiper motor and control module assembly
- Rear window washer pump
- Wiper arm and blade.

The rear window wiper and washer operation is controlled by the CJB. The rear window wiper operation is remote from the GEM control of the windshield wiper systems.

The rear window wiper motor and control module assembly is located in the upper tail door, behind a trim panel. The assembly is secured to a bracket on the upper tail door with 3, Torx head screws. Rubber bushes isolate the motor assembly from the bracket, which help reduce the transmission of motor operating noise to the tail door.

The assembly comprises the motor and the mounting bracket and the control module. The motor is located on a worm drive gearbox mechanism, which converts the rotary motion of the motor output spindle into the required arc for the rear window wiper blade. The control module is integral with the motor and has 2 plug sockets, which accept 2 multiplugs from the harness. One multiplug has 2.5 mm diameter wires. Two of the wires supply permanent battery feed and ground connections for the motor and control module. The third wire in this plug supplies a power supply to the rear window washer pump. The second multiplug has 0.35 mm diameter wires and carries signal data from the wash/wipe control stalk switch and the CJB. There is also a power input from the RJB (rear junction box).

The rear window washer feed hose is located at the rear of the motor spindle. The hose is connected to a 90° connector allowing the washer fluid to flow through the center of the motor spindle.

The motor spindle is a conventional design with a taper spline location for the wiper arm and a threaded shank to secure the arm to the spindle.

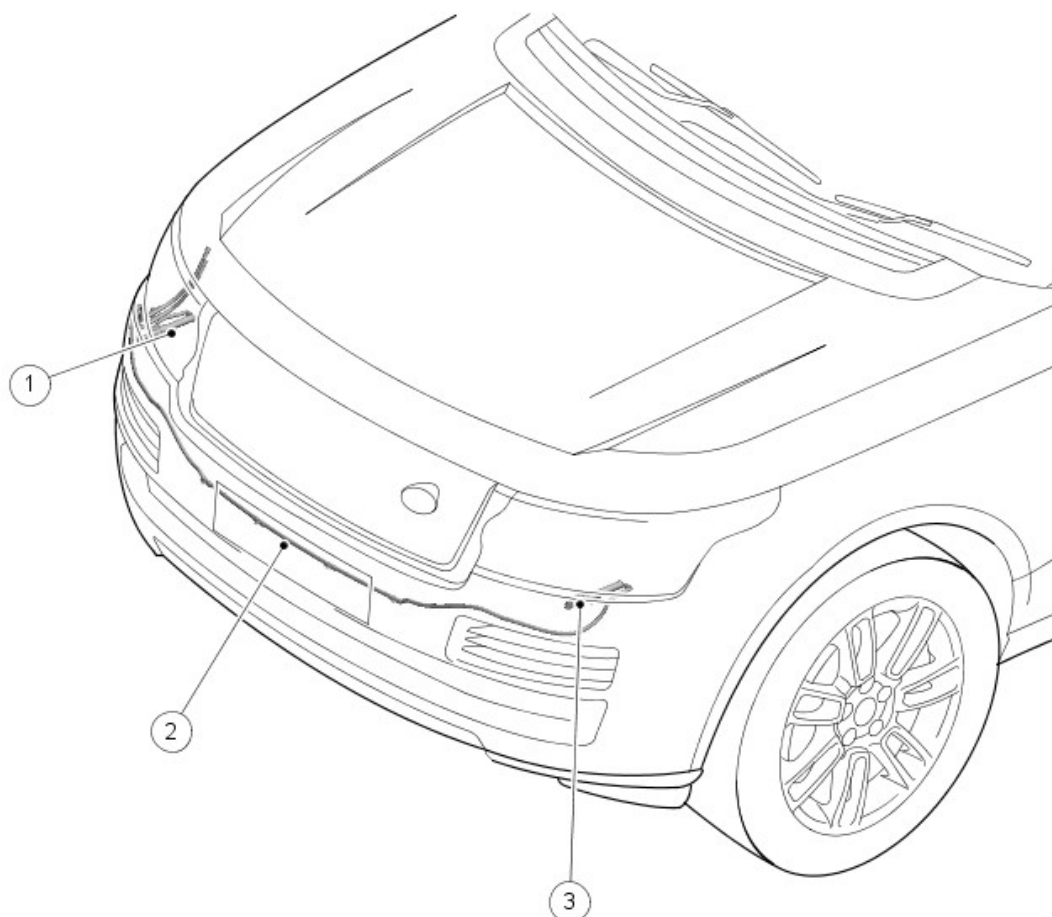
WIPER ARM

The wiper arm is similar in design to the windshield wiper arms. The arm attachment hole has tapered splines, which mate with the splines on the wiper spindle. The arm is secured to the wiper motor spindle with a spring washer and nut. The wiper arm has a pivot point, close to the spindle attachment. A tension spring is connected to the wiper arm on each side of the pivot point and applies pressure to maintain the wiper blade in contact with the windshield.

The wiper blade is attached to the wiper arm with a clip that allows the blade to pivot. The blade comprises a number of levers and yokes to which the rubber wiper blade is attached. The levers and yokes ensure that the pressure applied by the arm tension spring is distributed evenly along the full length of the blade and also allow the blade to adjust to the curvature and contour of the windshield.

A plastic cap located on the arm pivot point, covers the spindle attachment nut. The cap also contains 2, adjustable ball jets, which direct washer fluid onto the tail door window. The cap, when installed, connects with the hollow motor spindle, through which the washer fluid flows to the jets.

HEADLAMP WASHER SYSTEM



E149115

ITEM	DESCRIPTION
1	Right headlamp washer jet
2	Washer fluid hose
3	Left headlamp washer jet

A wash jet for each headlamp is located in a housing on the top surface of the front bumper. The jets are fed with fluid at high pressure from the headlamp wash pump. A large diameter hose connects each jet to the pump. Each connection is secured with a metal clip to secure the hose due to the high pressure from the pump. Each jet directs the high-pressure fluid in a wide spray onto the headlamp lens.

OPERATION

NOTES:

- If the hood is open windshield wiper operation is inhibited, although the wipers will still function with the windshield washers.
- If the tailgate is open the rear window wiper and washer systems are inhibited.

WINDSHIELD WIPERS

The windshield wiper system has the following functionality:

- Intermittent wipe and delay adjustment or rain/light sensor
- Normal (slow) speed wipe
- Fast speed wipe
- Program wash/wipe
- Flick wipe
- Headlamp power wash
- Motor blocking protection.

INTERMITTENT

Intermittent wiper operation is selected on the wiper control switch by moving the switch vertically to the first position. The intermittent delay period is adjustable using the rotary switch on the wiper control switch. The delay period is also subject to vehicle speed, with the selected delay period decreasing with an increase in road speed.

The rotary potentiometer selects differing resistance values for each position. The potentiometer is connected into a voltage divider circuit, with a 6.8kohm resistor. The power supply to the voltage divider circuit is limited to 6.5V. This reduced voltage is used to determine the position of the rotary

switch as follows:

ROTARY SWITCH POSITION	RESISTANCE	MIN. VOLTAGE	MAX. VOLTAGE	TYPICAL VOLTAGE
Error - Ground	<0.2kohm	-	-	<0.18V
1	2kohm $\pm$ 750ohm	1.00V	1.89V	1.48V
2	4kohm $\pm$ 750ohm	2.08V	2.70V	2.41V
3	6kohm $\pm$ 750ohm	2.80V	3.27V	3.04V
4	8kohm $\pm$ 750ohm	3.32V	3.69V	3.51V
Error - Positive	>20kohm	-	-	4.88V

The intermittent delay is also influenced by the road speed of the vehicle using a signal value derived from the Anti-Lock Brake System (ABS) control module on the high speed CAN (controller area network) bus. The time delay periods for the vehicle stationary and when moving at different speeds are shown in seconds in the following table:

ROTARY SWITCH POSITION	VEHICLE SPEED MPH (KM/H)						
	4 (6)	4 - 19 (6 - 30)	19 - 44 (30 - 70)	44 - 62 (70 - 100)	62 - 87 (100 - 140)	>87 (140)	>112 (180)
Error - Ground	20	6	5	4	3	3	3
1	26	19	17	15	15	13	13
2	17	12	11	10	9	7	7
3	10	6	6	5	4	3	3
4	5	3	3	2	2	2	2
Error - Positive	8	6	5	4	3	3	3

The wiper control switch positions also influence the operation of the rain/light sensor by adjusting its sensitivity. Refer to the following rain/light

sensor operation section for details.

NORMAL (SLOW) SPEED

The normal (slow) speed continuous wiper operation is selected by moving the switch vertically to the second detente position. The wipers will operate continuously when the vehicle is moving. When the vehicle is stationary (less than 4 mph (6 km/h)), the CJB operates the wipers in the intermittent mode, using a 3 second intermittent delay period.

FAST SPEED

The fast speed continuous wiper operation is selected by moving the switch vertically to the third detente position. The wipers will operate continuously at fast speed when the vehicle is moving. When the vehicle is stationary (less than 4 mph (6 km/h)), the CJB operates the wipers in normal (slow) speed mode.

RAIN/LIGHT SENSOR

The rain/light sensor is active when the wiper control switch is in the intermittent position. The rain/light sensor suspends wiper operation when the area of the windshield for the rain/light sensor is dry and operates the wipers continuously (fast or slow speeds) when the windshield is subject to heavy rainfall.

If the ignition is switched off and the wiper control switch remains in the intermittent position, the CJB will suspend rain/light sensor operation until intermittent operation is reselected, an adjustment of the rotary switch is made or programmed wash/wipe is selected.

The sensitivity of the rain/light sensor can be adjusted by the driver using the intermittent rotary switch on the wiper control switch. Six sensitivity levels of the sensor can be selected which has the effect of increasing or decreasing the wiper delay period, allow driver adjustment for the prevailing conditions. When several continuous wipe cycles have taken place, the sensor will maintain the continuous operation to avoid switching

back to intermittent from a continuous wipe and back again.

The rain/light sensor receives vehicle speed information from the ABS (anti-lock brake system) control module on the high speed CAN bus every 2 seconds. The sensor increases the sensitivity as the speed increases to optimize wiper operation. When the vehicle speed is reduced to less than 5 mph (8 km/h), the sensitivity is automatically reduced. Below this speed the wipers will only operate continuously in very heavy rain.

WASH/WIPE

Moving either the wiper control switch backwards selects windshield wash/wipe. This position is non-latching and the windshield washer pump is operated for as long as the switch is active. If the switch is only momentarily operated (less than 300 ms), only the washer pump operates. If the switch is activated for more than 300 ms, the CJB also powers the wiper motor after a 0.5 second delay. The CJB monitors the wash/wipe switch operation and after the switch is released, the CJB allows 3 full wipe cycles to clear the screen.

WASHER FLUID LEVEL SENSOR

The sensor cannot determine the precise amount of fluid in the reservoir, but can detect when the fluid level has fallen below a certain point. When the fluid level is low, the magnetic float closes the sensor contacts, completing a circuit through the sensor. This completed circuit is sensed by the CJB to which the sensor is directly connected.

The fluid level sensor is monitored continuously by the CJB. The CJB checks the fluid level sensor when the ignition is switched on to give the driver an early warning of the low fluid level. The CJB then monitors the sensor value over a 25 second period when the ignition is on to prevent invalid messages due to fluid 'sloshing' in the reservoir.

When the CJB determines a low fluid level, a signal is transmitted on the high speed CAN bus to the IC to illuminate the 'WASHER FLUID LOW' message in the IC message center. The first illumination of this message is accompanied by a 'gong' sound to alert the driver to the low fluid level.

HEATED WASHER JETS

The heated washer jets are controlled by the ATCM (automatic temperature control module) and a heated washer jet relay in the EJB.

The ATCM monitors the ambient air temperature broadcast by the ECM (engine control module) on the When the ignition is switched on and the ATCM detects an ambient temperature of less than 3°C (37°F), the ATC (automatic temperature control) system completes an earth path for the coil of the heated washer jet relay. This allows battery voltage to pass through the relay contacts and operate the heater elements. The ATC maintains the power to the heater elements for as long as the ignition remains on or the ambient temperature remains below 6°C (42°F). If the ambient temperature increases to 6°C (42°F) or above, the ATC system removes the earth for the relay coil, de-energizing the heater elements.

The ATC system will also prevent heated washer jet operation if the battery voltage falls to below 11.4V. This is to relieve loads on the battery and alternator when electrical loads are high. Heated washer jet operation will be resumed when the battery voltage exceeds 12.2V.

WIPER MOTOR BLOCKING PROTECTION

The wiper park signal is also used by the CJB for blocking protection of the windshield wiper motor. This feature protects the motor in the event of the wiper operation being obstructed.

If the CJB does not receive the wiper park signal for a period of 16 seconds, when the wiper motor is active, the CJB removes the power supply to the motor. The motor is disabled for a period of 3 minutes or until the ignition is switched off and on again. Even after the 3 minute period has elapsed, the CJB will not automatically switch the motor on, to prevent the risk of injury. The wiper switch must be moved off and then on to reactivate the wiper motor. The blocking protection is active in all wiper switch positions and can only be reset by turning the ignition off.

REAR WINDOW WIPER

The rear window wiper has the following functionality:

- Intermittent operation
- Adjustable intermittent wipe period
- Continuous reverse wipe function.

WIPER OPERATION

The rear window wiper is operated by the wiper control switch. Rotating the outer ring of the switch the first position switches on the rear window wiper intermittent mode. Rotating the switch forward to the second position switches on the rear window wiper normal mode. Pressing the switch end selects the rear window wash/wipe function.

When reverse gear is engaged and the rear window wiper is on, the rear window wiper is operated continuously. The continuous wipe will continue until reverse gear is disengaged or the rear window wiper is switched off.

When rear window wash/wipe is selected, the EJB powers the washer pump for as long as the switch is held. When the switch is released, the EJB initiates 5 complete wipe cycles. The first 3 wipe cycles are each accompanied by a 0.5 second operation of the rear window washer pump. The final 2 cycles are to clear the tail door window and do not have a washer pump operation. If intermittent remains selected, the rear window wiper continues in intermittent mode.

INTERMITTENT ADJUSTMENT

The intermittent interval time of the rear window wiper can be adjusted by the driver to wipe at between 1 and 30 second delays. Adjustment is performed by switching the rear window wiper on for a short time and then off again. The next switching on of the rear window wiper determines the delay period. The time lapse between the off and the next on signal is sensed by the CJB which sets this period as the requested time interval between wipes. If the next on signal is more than 30 seconds after the off signal, the CJB sets the time interval to the maximum 30 seconds.

Rear window wiper Motor Blocking Protection

The rear window wiper motor is protected by the CJB in the event of the wiper operation being obstructed.

HEADLAMP WASHER

The headlamp washers are only active when the headlamps are switched on and the ignition is on. The washers are controlled by the CJB, which limits their operation to preserve washer fluid in the reservoir.

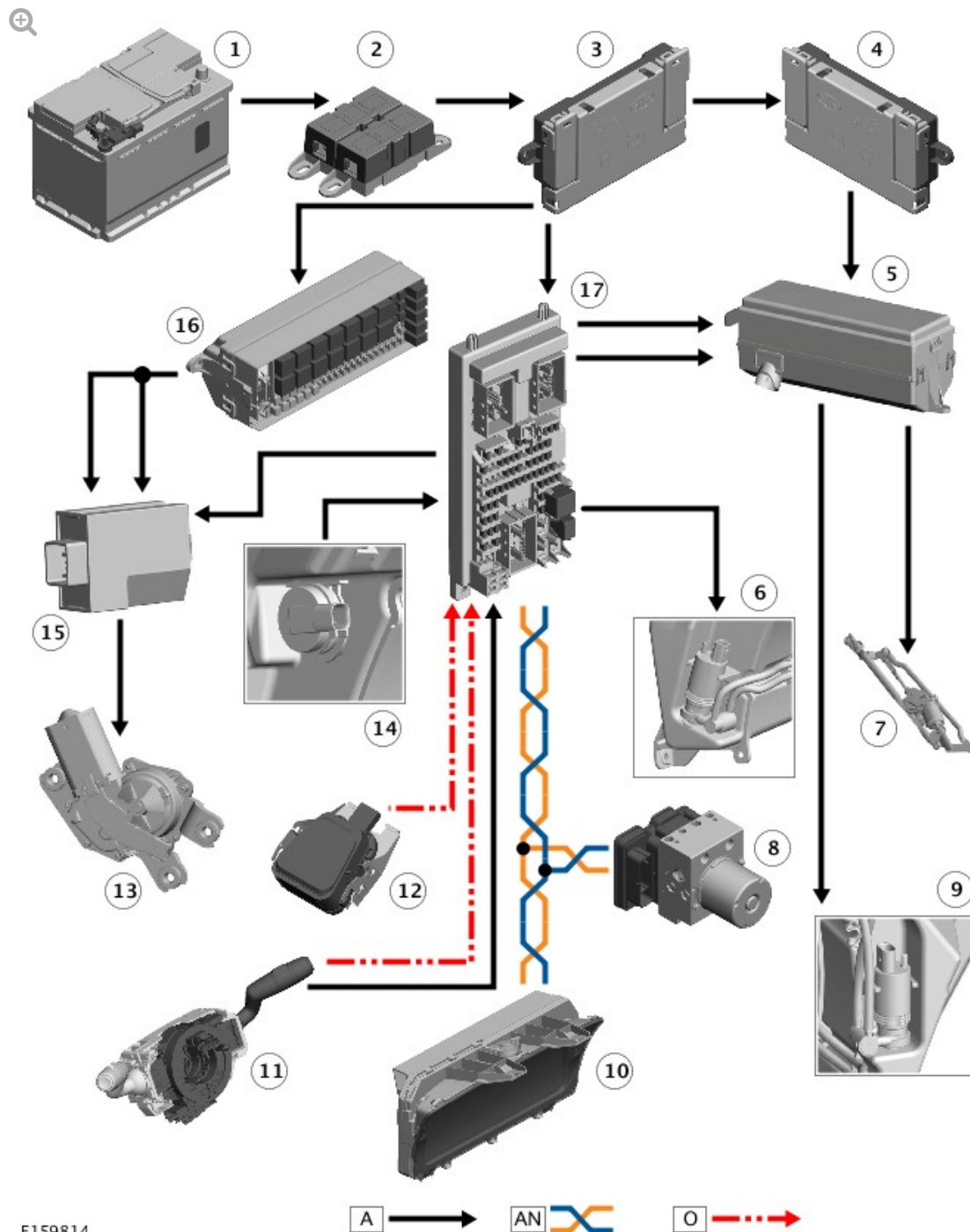
With the ignition and lights on, the headlamp wash is activated on the first operation of the wiper control switch in the wash/wipe position. The CJB then suspends headlamp wash activation for the next 4 operations of the wash/wipe switch, with the headlamp power washers activated on the fifth operation of the switch.

The CJB monitors the operation of the wash/wipe switch and maintains a counter to restrict headlamp power wash operation to every fifth operation of the wash/wipe switch. The CJB also maintains a timer, which prevents a second operation of the headlamp wash within a 3 minute period. The counter and timer are reset when the ignition is switched off.

When headlamp power wash is active, the CJB energizes the wash pump twice per cycle. The headlamp wash pump is powered for a 0.5 second period. The CJB then initiates a 2 second delay before powering the headlamp wash pump for a second 0.5 second period.

If the washer fluid level becomes low, the CJB suspends headlamp power wash operation to preserve the remaining washer fluid.

CONTROL DIAGRAM - WIPERS AND WASHERS

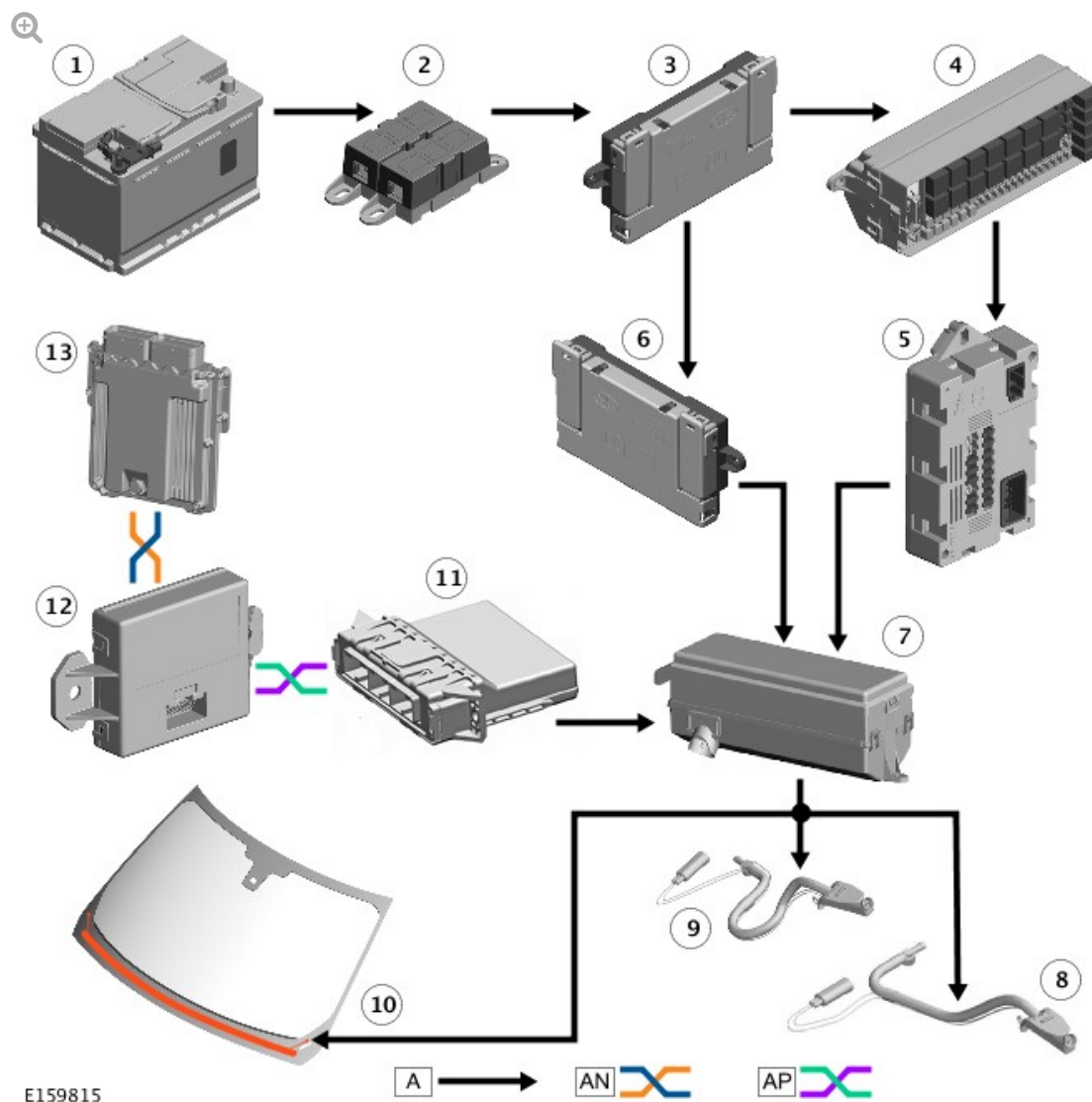


A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; O = LIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB) 2
3	BJB
4	Auxiliary Junction Box (AJB)

5	Engine Junction Box (EJB)
6	Windshield/Rear window washer pump
7	Windshield wiper motor
8	Anti-Lock Brake System (ABS) control module
9	Headlamp washer pump
10	Instrument Cluster (IC)
11	Wipers and washers control switch and clockspring
12	Rain/Light sensor
13	Rear window wiper motor
14	Washer fluid level sensor
15	RF filter
16	Rear Junction Box (RJB)
17	Central Junction Box (CJB)

CONTROL DIAGRAM - HEATED WASHER JETS



A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; AP = HIGH SPEED CAN CHASSIS SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	BJB 2
3	BJB
4	RJB
5	Quiescent current control module
6	AJB
7	EJB
8	Left heated washer jet

9	Right heated washer jet
10	Wiper blade heater
11	Automatic Temperature Control Module (ATCM)
12	Gateway Module (GWM)
13	Engine Control Module (ECM)